David Alexander Machado Alzate

Backend Software Engineer · .NET & TypeScript

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SUMMARY

Software engineering student (5th semester, Universidad Pascual Bravo) focused on backend systems: transactional APIs, relational schemas, and domain logic covered by automated tests. Built and documented a .NET 10 marketplace core with hexagonal architecture, DDD, CQRS, and 592 xUnit tests, plus an open-source TypeScript lease billing engine with a Vitest suite. Looking for a backend team where I can review code, own small services, and learn from production systems. I work best where documentation, clear communication, and code review are part of the routine.

EXPERIENCE

Independent Software Developer 2024 – Present

Client projects & open source

- **Zentric — Marketplace Core API:** designed the domain core of a multi-vendor marketplace in .NET 10 with ports and adapters, DDD aggregates, and a CQRS split of 22 command and 14 query handlers. Mapped 14 DbSets with EF Core and PostgreSQL, containerized the service with Docker, documented decisions as ADRs, and shipped 592 xUnit tests across domain, application, and API layers.
- **LeaseFlow — Lease Billing Engine:** built an open-source TypeScript engine with Express and SQLite that turns rent, partial payments, and receipts into deterministic balances. Wrote the Vitest suite covering domain money rules and the HTTP layer.

SELECTED PROJECTS

Zentric — Marketplace Core API github.com/D-MachadoDev/Zentric

.NET 10 · Hexagonal Architecture · DDD · CQRS · PostgreSQL · Docker · 592 tests

LeaseFlow — Lease Billing Engine github.com/D-MachadoDev/leaseflow

TypeScript · Express · SQLite · Vitest · partial payments and reconciliation

Retail Analytics — Data Pipeline & BI Private case study

Incremental ETL · star schema · Power BI dashboards · walkthrough on request

SKILLS

Languages	C#, TypeScript, SQL, Python
Backend	.NET 10, Express, REST APIs
Data	PostgreSQL, SQLite, EF Core
Architecture	Hexagonal Architecture, DDD, CQRS, SOLID
Testing	xUnit, Vitest
Tooling	Docker, Git, GitHub Actions
Working style	Written communication, ownership, self-directed learning, collaboration

EDUCATION

Universidad Pascual Bravo — Ingeniería de Software 2022 – Present · 5th semester

Coursework: object-oriented programming, software architecture, relational databases, data structures, operating systems.